RAHUL TIWARI

AI-Native Backend Engineer | Java | Spring Boot | GenAI | RAG | MCP | Microservices
Pune, IN | tiwarirahul0809@gmail.com |linkedin.com/in/rahultiwaridev| PORTFOLIO | GITHUB

SUMMARY

AI-Native Backend Engineer with 3+ years building **fault-tolerant Java/Spring Boot microservices** for Citi's branded cards platform, and **production GenAI: Spring AI, RAG, agents, MCP and LLM evaluation**. Skilled in system design, Kafka, database tuning and CI/CD. AWS Certified Solutions Architect and **4× Claude Certified** (incl. Architect – Professional).

WORK EXPERIENCE

Software Engineer | Citi (via Hexaware Technologies) *Chennai, IN | 08/2024 – Current*

- Built card-processing and transaction microservices for the branded cards platform, sustaining **1,200 RPS** with retries and fallbacks that keep it available during partial outages.
- Built an **AI-powered PR review pipeline** for the team that parses Git diffs and posts inline fix suggestions, speeding up code reviews by **32%** and cutting syntax bugs by **18%**.
- Developed and enhanced **80+ REST APIs** across **20+ microservices** with **Swagger/OpenAPI** for cross-team integration; cut downstream calls per request on 20+ APIs (e.g., 4 → 2) by enriching response payloads, lowering end-to-end latency.
- Delivered a data migration framework moving **3.5M financial accounts** at **99.9% accuracy** with zero downtime.
- Engineered a Recovery-as-a-Service (RaaS) framework with state tracking and exponential backoff, cutting timeouts by **35%** and auto-recovering **88%** of timed-out inter-service calls.
- Built an async Kafka pipeline for HRT streams with **sub-55ms anomaly detection** via partition and consumer-group tuning.
- Added distributed Redis caching and restructured SQL indexes, cutting retrieval latency by **26.3%** and DB CPU by **22.5%**.
- Raised code coverage to **90%+** with TDD and an automated BDD (Gherkin) pipeline, cutting manual QA effort by **60%**.
- Built Splunk dashboards and logging that cut P1/P2 MTTR by **48%**; own root-cause analysis and monthly CI/CD releases.

Associate Software Engineer | Hexaware Technologies — Enterprise LMS *Chennai, IN | 09/2023 – 07/2024*

- Built REST services and **event-driven notifications** for content allocation and learning tracking; sped up hot endpoints with pagination, indexing and caching.
- Improved frontend performance (Lighthouse **+24%**, bundle **−140KB**); built an AI onboarding chatbot deflecting **38%** of tier-1 tickets.

PROJECTS

AI PR Reviewer *Java 25, Spring Boot 4, Spring AI, PostgreSQL/pgvector, Kafka*

- Built a **GitHub App that reviews PRs with an LLM** (Gemini/OpenAI via config) and **posts inline comments**; a custom diff parser maps model-chosen locations to real lines, so every comment lands on a valid line.
- **Engineered duplicate-safe publishing** (HMAC-verified webhooks, delivery-ID dedup, hidden markers, head-commit checks) so redeliveries, lost responses and new pushes never cause duplicate or stale reviews.
- **Hardened against prompt injection** with a tool-less model and Java validation of every finding; backed by **162 network-free tests** in CI and **20 ADRs**.

PEN.AI (Document AI SaaS Platform) GITHUB | LINK

- Full-stack Next.js SaaS for **chatting with PDFs** (Prisma, PostgreSQL, Stripe), supporting **450+ concurrent sessions**.
- **RAG pipeline**: Uploadthing (S3) uploads embedded into Pinecone, with **~120ms** vector-search retrieval.

Board (Real-Time Collaborative Whiteboard) GITHUB | LINK

- Next.js 14 multiplayer whiteboard (shapes, notes, freehand) at **60 FPS for 50+ concurrent users** over WebSockets.
- Built drawing tools from scratch as **non-blocking vector tools**; Clerk auth, Convex real-time DB, shadcn/ui.

CERTIFICATIONS

- AWS Certified Solutions Architect – Associate (SAA-C03)
- **Anthropic (4/4):** Claude Certified Architect – Professional & Foundations · Developer – Foundations · Associate – Foundations

EDUCATION

B.Tech in Electronics & Communication | LNCTE, Bhopal | CGPA 7.9 *Bhopal, IN | 07/2019 – 03/2023*

SKILLS

Languages & Frameworks: Java, TypeScript, Python, Spring Boot, Spring AI, Node.js, Express, React, Next.js, Redux, Tailwind CSS
GenAI & Agents: Agent Orchestration (LangChain, LangGraph, Google ADK, LiteLLM), Model Context Protocol (MCP), RAG & GraphRAG, Prompt Engineering, Fine-tuning, LLMs (AWS Bedrock, Azure OpenAI, Anthropic Claude API), LLM Evaluation
Data & Messaging: SQL, PostgreSQL (pgvector), MySQL, MongoDB, Redis, Pinecone, Kafka
Cloud, DevOps & Testing: AWS (EC2, RDS, ELB, S3), Docker, GitHub Actions, Jenkins, Splunk, Swagger/OpenAPI, JUnit 5, Testcontainers, BDD (Gherkin)

OTHER

Best Team of the Year (2024) Awarded for exceptional team performance in delivering high-impact projects.
Designathon (2025) Awarded 2nd place in the Hexaware Designathon.